South Campus Sensor Network: Gateway Placement Study

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1 Introduction

This note documents the proposed placement of wireless gateways for the south campus environmental sensor network. Each building in the pilot area carries a pair of field probes—one temperature probe and one humidity probe—that report over a low-power radio link to a building-level gateway. The gateways forward readings over the wired backbone to a central collector, which archives them for the building management system.

The pilot covers twelve buildings, but the topology repeats: every building instantiates the same four-node pattern shown in Figure 1. The only per-building parameters are the radio distances, which range from 8 to 40 metres, and the reporting interval, which is five minutes for temperature and fifteen minutes for humidity.

2 Network Layout

Figure 1 shows the reference pattern. The two field probes sit at the top; the wavy edges denote the wireless hop to the gateway, drawn here with a snake decoration to distinguish it from the wired links. The gateway aggregates both streams and forwards them to the collector over Ethernet. The dashed outline groups the field cluster, while the shaded backdrop marks the equipment that lives in the building wiring closet.

The midpoint construction in the diagram is deliberate: the gateway is centred horizontally between its two probes, so the radio diagram stays symmetric whatever the probe spacing. Distances were surveyed in September; the longest wireless hop is 38 metres through two interior walls, which bench tests show still leaves 12 dB of link margin.

3 Findings

The study recommends one gateway per building, mounted in the wiring closet beside the collector uplink. The measured packet delivery rate over two weeks was 99.7 %, with all losses confined to a single gateway whose antenna was later found misaligned. Re-aiming the antenna cleared the errors, and no losses have been observed since.

Total hardware cost for the pilot is twelve gateways plus the collector VM, well inside the project budget. The next step is to extend the same pattern to the north campus residences, where the longer inter-building distances may require an additional relay hop; that extension will be the subject of a follow-up note.

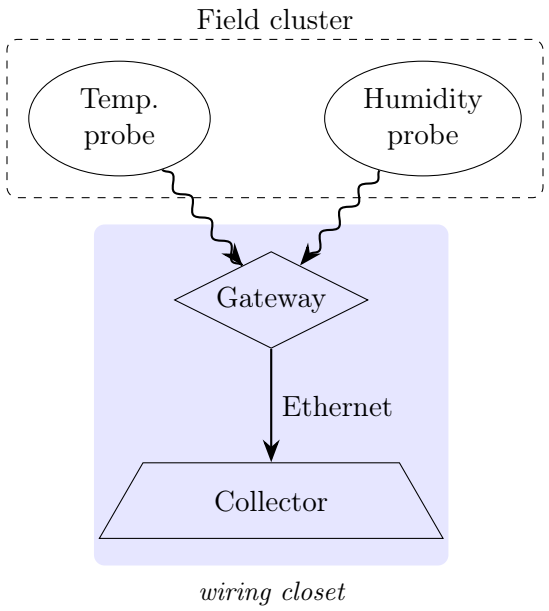


Figure 1: Reference four-node topology, repeated in every pilot building.